Discipline: Adaptive and Robust Control

Exam tasks/questions

Task №	Description
1	Solve the problem of **adaptive** control of the plant. Find the control u that will provide the boundedness of all the closed—loop signals and the convergence of the plant **output** to the output of the **reference model:** $\lim_{t\to\infty}(\varepsilon(t))=0$, where $\varepsilon(t)=y_m(t)-y(t)$. Simulate, make plot of $\varepsilon(t),u(t)$
2	Solve the problem of **adaptive** control of the plant. Find the control u that will provide the boundedness of all the closed—loop signals and the convergence of the plant **state** to the state of the **reference model:** $\lim_{t\to\infty}(e(t))=0$, where $e(t)=x_m(t)-x(t)$. Simulate, make plot of $e(t),u(t)$.

General notations: θ_i,θ are unknown constant parameters; $\tilde{\theta}_i=\theta_i-\hat{\theta}_i$, $\tilde{\theta}=\theta-\hat{\theta}$ are the parametric errors; $\hat{\theta}_i,\hat{\theta}$ are the estimates of θ_i,θ (for identification problems) or the adjustable parameters (for control problems), g is a bounded piece-wise continuous reference signal.

Initial data for simulations (choose your parameters depending on the variant)

Unknown parameters: $\theta_1=1,\theta_2=2,\theta_3=3,\ \theta=4$**;**

Known threshold for projection: $\theta_0=0.1$**;**

Reference signal: $g(t)=N\sin t+(N-2)$, N *is the number of variant* (*the first column*)

Initial conditions in the adaptation algorithm: $\hat{\theta}_1(0)=\hat{\theta}_2(0)=\hat{\theta}_3(0)=\hat{\theta}(0)=0.5$

	Name	Plant	Reference model	Task №
1.	LI MINGSI	$\dot{y}=\theta_1 y^2+\theta_2 y+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**
2.	CHEN YIQIANG	$\dot{y}=\theta_1 y+\theta_2 y^3+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**
3.	XIONG XIAOLIN	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1 \sin x_1+\theta_2 x_2+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-x_{m1}-x_{m2}+g\end{cases}$	**2**
4.	LU CHENXI	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta e^{x_1}+(1+x_1^2+x_2^2)u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-x_{m1}-x_{m2}+g\end{cases}$	**2**
5.	GUO YUYOU	$\dot{y}=\theta_1 y^3+\theta_2 \sin y+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**
6.	LI XIN	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1 \cos x_1^2+\theta_2 \sin x_2^2+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-x_{m1}-x_{m2}+g\end{cases}$	**2**
7.	ZHOU XINGYU	$\dot{y}=\theta_1 \sin(y)+\theta_2 sign(y)+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**
8.	ZHANG CHENXI	$\dot{y}=\theta_1+\theta_2 sign\left(\sin\ y\right)+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**
9.	XU HAIPENG	$\dot{y}=-\cos y+\theta(y^2+1)u,$ $\theta \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-5y_m+5g$	**1**
10.	WU YANGNIAN	$\dot{y}=\theta_1 e^y+\theta_2 y^3+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**
11.	ZHANG LETIAN	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1 x_1^3+\theta_2+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-x_{m1}-x_{m2}+g\end{cases}$	**2**
12.	NING BO	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=(1+\theta_1 x_1^2+\theta_2 x_2^2)+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-6x_{m1}-7x_{m2}+6g\end{cases}$	**2**
13.	KONG FANLIN	$\dot{y}=\theta_1 e^{-y}+\theta_2 u,$ $\theta_2 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-7y_m+7g$	**1**
14.	LI YUANXIAO	$\dot{y}=\theta_1 \cos y^3+\theta_2 y+\theta_3 u,$ $\theta_3 \geq \theta_0>0,\ \ \theta_0$ *is known*	$\dot{y}_m=-y_m+g$	**1**

15.	LI JINGSEN	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1 x_1^2 + \theta_2 x_2^2 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -7x_{m1} - 4x_{m2} + 7g\end{cases}$	**2**
16.	OUYANG JUNZHE	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1 - \theta_1 x_1^2 - \theta_2 x_2^2 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1} - x_{m2} + g\end{cases}$	**2**
17.	HUANG ZHENNAN	$\dot{y} = -\theta_1 y^3 - \theta_2 y + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -2y_m + 2g$	**1**
18.	WANG SANLI	$\dot{y} = \theta_1 \ln(1+y^2) + \theta_2 u,$ $\theta_2 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -y_m + g$	**1**
19.	ZHANG WEI	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta_1 sign(x_1) - \theta_2 x_2 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g\end{cases}$	**2**
20.	CHEN BOXUAN	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1 \cos(x_1^2) + \theta_2 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1} - 3x_{m2} + g\end{cases}$	**2**
21.	LIANG SHUANG	$\dot{y} = \dfrac{\theta_1}{2+\sin y} + \theta_2 u,$ $\theta_2 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -7y_m + 7g$	**1**
22.	JIANG JUNYU	$\dot{y} = \theta_1 \sin y + \theta_2 \cos y^2 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -7y_m + 7g$	**1**
23.	GUO YITONG	$\dot{y} = \theta_1 sign\left(y^3\right) + \theta_2 y + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -\lambda y_m + \lambda g,$ $\lambda > 0 \text{ is a constant}$	**1**
24.	SHI QIUXI	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta_1 sign\, x_1 - \theta_2 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1} - 3x_{m2} + g\end{cases}$	**2**
25.	YANG BOHAN	$\dot{y} = \theta_3 u + \theta_1 \sin t + \theta_2 \cos 2t,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -3y_m + 3g$	**1**
26.	YU JIAMIN	$\dot{y} = 2 + \theta_1 y^3 + \theta_2 y + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0 \text{ is known}$	$\dot{y}_m = -2y_m + 2g$	**1**
27.	HE JINSEN	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta \sin x_1 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -6x_{m1} - 5x_{m2} + 6g\end{cases}$	**2**
28.	ZHANG SHITONG	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1 \sin(x_1) + \theta_2 \cos(x_1) + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -8x_{m1} - 6x_{m2} + 8g\end{cases}$	**2**

29.	TAO ZHENG	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 x_2 \cos(x_1) + \\ \qquad + \theta_2 x_1 \sin(x_2) + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 3x_{m2} + g\end{cases}$	**2**
30.	ZHANG JIAXUAN	$\dot{y} = \theta_1 y^3 + \theta_2 \sin y^3 + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -4y_m + 4g$	**1**
31.	MENG ZEANG	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 x_1^3 + \theta_2 \left(x_1^2 - x_2^2\right) + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 3x_{m2} + g\end{cases}$	**2**
32.	ZHENG KAISA	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 \left(x_1^2 - x_2^2\right) - \theta_2 x_1 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -4x_{m1} - 4x_{m2} + 4g\end{cases}$	**2**
33.	YE LEI	$\dot{y} = -\theta_1 \sin y - \theta_2 y + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -4y_m + 4g$	**1**
34.	XU YAOWEN	$\dot{y} = \theta_1 \sin 3t + \theta_2 \sin 6t + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -3y_m + 3g$	**1**
35.	SUN NINGMAN	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = -\theta_1 - \theta_2 \left(x_1^2 + x_2^2\right) + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -3x_{m1} - 4x_{m2} + 3g\end{cases}$	**2**
36.	LEI JIALU	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = x_2^2 \left(\theta_1 + \theta_2 \sin(x_1)\right) + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -7x_{m1} - 8x_{m2} + 7g\end{cases}$	**2**
37.	WU XINYAO	$\dot{y} = -\theta_1 \sin y^2 + \theta_2 \cos y^2 + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -9y_m + 9g$	**1**
38.	PENG YINGRUI	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 + \theta_2 \sin(x_1^2 + x_2^2) + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -12x_{m1} - 7x_{m2} + 12g\end{cases}$	**2**
39.	ZHANG XUAN	$\begin{cases}\dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 \sin(x_1 x_2) + \theta_2 + u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -3x_{m1} - 2x_{m2} + 3g\end{cases}$	**2**
40.	LIN MINGHAO	$\dot{y} = \theta_1 y + \theta_2 y^2 + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -8y_m + 8g$	**1**
41.	LIANG JUNXI	$\dot{y} = \theta_1 y + (\theta_1 + \theta_2) y^3 + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -2y_m + 2g$	**1**
42.	TAO PENGYU	$\dot{y} = \theta_1 y \sin y + \theta_2 y + \theta_3 u,$ $\theta_3 \geq \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -4y_m + 4g$	**1**

43.	LIU RISHENG	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta\ln(1+x_1^2+x_2^2)-u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -4x_{m1}-5x_{m2}+4g\end{cases}$	**2**
44.	LI MINGXIANG	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta-(1+x_1^2+x_2^2)u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -4x_{m1}-5x_{m2}+4g\end{cases}$	**2**
45.	YANG ZHENHUA	$\dot{y} = \theta_1 y^2+\theta_2 y^3+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0$ *is known*	$\dot{y}_m = -5y_m+5g$	**1**
46.	SUN HAOZHOU	$\dot{y} = y^2+\theta\left(y^2+1\right)u,$ $\theta \ge \theta_0 > 0,$ θ_0 *is a known constant*	$\dot{y}_m = -y_m+g$	**1**
47.	ZHANG WEI	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta_1-\theta_2(1+x_1^2)+(2+\sin x_2)u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -2x_{m1}-3x_{m2}+2g\end{cases}$	**2**
48.	CHEN SHITAO	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta_1(1+x_2^2)-\theta_2(1+x_1^2)+u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1}-5x_{m2}+g\end{cases}$	**2**
49.	BIAN XIJIE	$\dot{y} = \theta_1 y^4+\theta_2\left(y^2+1\right)u,$ $\theta_2 \ge \theta_0 > 0,\ \ \theta_0$ *is known*	$\dot{y}_m = -3y_m+3g$	**1**
50.	LI GUOWEN	$\dot{y} = \left(\theta_1+\theta_2\right)y+\theta_2 y^3+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0$ *is known*	$\dot{y}_m = -2y_m+2g$	**1**
51.	ZHAO MINZHENG	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = -\theta_1-\theta_2\sin x_2+(1+x_1^2)u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1}-2x_{m2}+g\end{cases}$	**2**
52.	NIU XIAOXIN	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1+\theta_2\sin x_1+\theta_3\cos x_1+u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -3x_{m1}-4x_{m2}+3g\end{cases}$	**2**
53.	ZHANG SEN	$\dot{y} = \sin t+\theta\left(1+0.5y^2\right)u,$ $\theta \ \ge \theta_0 > 0,\ \ \theta_0$ *is known*	$\dot{y}_m = -2y_m+2g$,	**1**
54.	YANG LEI	$\dot{y} = \theta_1\left(1+y^2\right)^{-1}+\theta_2 y+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\ \ \theta_0$ *is known*	$\dot{y}_m = -4y_m+4g$,	**1**
55.	FU HAIYANG	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1 x_2+\theta_2 x_2^2+\theta_3 x_2^3+u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -x_{m1}-2x_{m2}+g\end{cases}$	**2**
56.	BAO PENGFAN	$\begin{cases}\dot{x}_1 = x_2,\\ \dot{x}_2 = \theta_1 x_1^2-\theta_2 x_1 x_2+\theta_3 x_2^2+u\end{cases}$	$\begin{cases}\dot{x}_{m1} = x_{m2},\\ \dot{x}_{m2} = -6x_{m1}-5x_{m2}+6g\end{cases}$	**2**

57.	ZHU CUNXI	$\dot{y} = -\theta_1 - \theta_2 y^3 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -7y_m + 7g$,	**1**
58.	JIANG HAOTIAN	$\dot{y} = \theta_1 \sin y + \theta_2 \cos y + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -3y_m + 3g$	**1**
59.	HE RUNMIN	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -\theta_1 (x_1 + x_2)^2 - \theta_2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -9x_{m1} - 6x_{m2} + 9g \end{cases}$	**2**
60.	WU MO	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**
61.	ZHOU YINAN	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = 1 - \theta_1 \lvert x_1 \rvert - \theta_2 \lvert x_2 \rvert + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -4x_{m1} - 4x_{m2} + 4g \end{cases}$	**2**
62.	WANG YINAN	$\dot{y} = \theta_1 \left(y^2 + 1 \right) + \theta_2 \left(y^2 + 1 \right) u,$ $\theta_2 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -y_m + g$	**1**
63.	ZHA MENGDU	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = (1 - \theta_1 x_1^2 - \theta_2 x_2^2) + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**
64.	ZONG RUIKANG	$\dot{y} = \theta_1 \lvert y \rvert + \theta_2 y + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -2y_m + 2g$	**1**
65.	DONG SIYA	$\dot{y} = \theta_1 y + \theta_2 y^3 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -y_m + g$,	**1**
66.	DING YAQI	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -\theta_1 (1 + x_1^2 + x_2^2) + \theta_2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**
67.	FENG QIRUI	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 \left(x_1^2 + 1 \right) + \theta_2 \left(x_2^2 + 1 \right) + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -3x_{m1} - 4x_{m2} + 3g \end{cases}$	**2**
68.	MAO SHIQI	$\dot{y} = \theta_1 y^3 + \theta_2 y^6 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -y_m + g$,	**1**
69.	CAI XUANHAO	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 + \theta_2 x_1^2 + \theta_3 x_2^2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**
70.	LIU FENG	$\dot{y} = \theta_1 \tanh y - \theta_2 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -4y_m + 4g$	**1**
71.	XU ZIYANG	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \left(\theta_1 x_1^2 + \theta_2 \right) x_2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 4x_{m2} + g \end{cases}$	**2**

72.	JIANG SHENGNAN	$\dot{y} = -\theta_1 - \theta_2\left(1 + y^3\right) + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -2y_m + 2g$	**1**
73.	WANG YUJIE	$\dot{y} = \theta_1 y \sin t + \theta_2 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -6y_m + 6g$,	**1**
74.	ZHAN SHUO	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 + \theta_2 x_1 + \theta_3 x_2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**
75.	LU LIN	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \cos x_1 \left(\theta_1 + \theta_2 \sin x_1\right) + \theta_3 u, \\ \theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known} \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -2x_{m1} - 3x_{m2} + 2g \end{cases}$	**2**
76.	ZHANG SHUYUAN	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 \sin x_2 - \theta_2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -8x_{m1} - 6x_{m2} + 8g \end{cases}$	**2**
77.	BIE ZHI	$\dot{y} = \theta_1 \sin t + \theta_2 y \sin t + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -\lambda y_m + \lambda g$, λ is a constant	**1**
78.	CHEN DONGREN	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \left(1 + \cos x_1\right)\left(\theta_1 + \theta_2 x_1^3\right) + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -8x_{m1} - 6x_{m2} + 8g \end{cases}$	**2**
79.	YUAN JIA	$\dot{y} = \theta_1 + \theta_2\left(1 - \sin y\right) + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -3y_m + 3g$	**1**
80.	ZHANG MIAOTE	$\dot{y} = -\theta_1 y - \theta_2\left(1 + \sin y\right) + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -y_m + g$	**1**
81.	DAI JIN	$\dot{y} = \theta_1 + \theta_2\left(1 + e^y\right) + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -2y_m + 2g$	**1**
82.	XUE HAORAN	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 x_1^2 + \theta_2 x_1 x_2 + \theta_3 x_2^2 + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**
83.	SUN HAO	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \theta_1 x_1 + \theta_2 x_1 x_2 + u, \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -5x_{m1} - 6x_{m2} + 5g \end{cases}$	**2**
84.	YANG LANQIN	$\dot{y} = -\theta_1 \sin y - \theta_2 \cos y + y^2 + \theta_3 u,$ $\theta_3 \ge \theta_0 > 0, \quad \theta_0 \text{ is known}$	$\dot{y}_m = -y_m + g$	**1**
85.	XIAO YAO	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = -\theta_1 \sin x_1 - \theta_2 \sin(2x_1) - \theta_3 \sin(4x_1) + u \end{cases}$	$\begin{cases} \dot{x}_{m1} = x_{m2}, \\ \dot{x}_{m2} = -x_{m1} - 2x_{m2} + g \end{cases}$	**2**

86.	LI JINMING	$\dot{y}=\theta_1+y^5+\theta_2 u,$ $\theta_2 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-6y_m+6g$	**1**
87.	QIU GENGYI	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1 x_1 x_2^2+\theta_2 x_1^2 x_2+u,\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-5x_{m1}-6x_{m2}+5g\end{cases}$	**2**
88.	WENG JINYAO	$\dot{y}=-\theta_1-\theta_2 y^2-y^4+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-6y_m+6g$	**1**
89.	LIU ANGE	$\dot{y}=-(\theta_1 y+\theta_2 y^2)\sin y+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-y_m+g$	**1**
90.	MA NAN	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1+\theta_2\sin t+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-2x_{m1}-3x_{m2}+2g\end{cases}$	**2**
91.	YANG HAOYU	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1\sin x_1+\theta_2+\theta_3\sin t+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-5x_{m1}-6x_{m2}+5g\end{cases}$	**2**
92.	ZHU CHENHAO	$\dot{y}=\theta_1+\theta_2 y^3+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-2y_m+2g$	**1**
93.	ZHANG YICHONG	$\dot{y}=\theta_1\sin t+\theta_2\sin 0.5t+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-4y_m+4g$	**1**
94.	GONG BOYU	$\dot{y}=\theta_1+\theta_2\dfrac{1}{y^2+1}+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-3y_m+3g$	**1**
95.	XIONG ZHUOEN	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1 x_1 x_2+\theta_2 sign(x_2)+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-x_{m1}-2x_{m2}+g\end{cases}$	**2**
96.	YANG ZONGYUE	$\dot{y}=\theta_1\sin\left(y+y^3\right)+\theta_2+\theta_3 u,$ $\theta_3 \ge \theta_0 > 0,\quad \theta_0 \text{ is known}$	$\dot{y}_m=-y_m+g$	**1**
97.	HOU SHENGYE	$\begin{cases}\dot{x}_1=x_2,\\ \dot{x}_2=\theta_1 x_2+\theta_2 x_1+u\end{cases}$	$\begin{cases}\dot{x}_{m1}=x_{m2},\\ \dot{x}_{m2}=-6x_{m1}-5x_{m2}+g\end{cases}$	**2**